Food System

En analys från Swedish Subsidiarity

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Food Systems: The Sovereignty Paradox

The Efficiency Paradox

Sweden has optimized its food system into fragility. A nation with significant agricultural capacity imports approximately 50% of its food through supply chains controlled by three retail corporations, processing infrastructure concentrated in a handful of facilities, and monoculture production that has eliminated the diversity required for resilience. This is not accidental inefficiency—it is the predictable outcome of applying industrial efficiency logic to complex adaptive systems.

The paradox: Sweden achieves apparent efficiency (low prices, consistent availability) by creating brittleness (import dependency, single points of failure, loss of adaptive capacity). This represents a category error in system design: optimizing for one variable (cost per unit) while ignoring systemic properties (resilience, sovereignty, requisite variety).

Active System Failures

Monoculture and Fragility: The Systemic Risk of Centralization

The mechanism: 25% of farms generate 80% of agricultural value. Monoculture cereals (primarily wheat and barley) occupy roughly 40% of arable land. This concentration creates correlation: when conditions fail for one, they fail for many. The system has eliminated the redundancy and diversity that enable adaptation.

The cybernetic failure: A complex, variable environment (Swedish agriculture across diverse climates, soils, and markets) governed by a low-variety controller (standardized industrial methods). This violates Ashby's Law of Requisite Variety: only variety can absorb variety. Centralized monoculture cannot match the complexity of local conditions, making system-wide failure inevitable under stress.

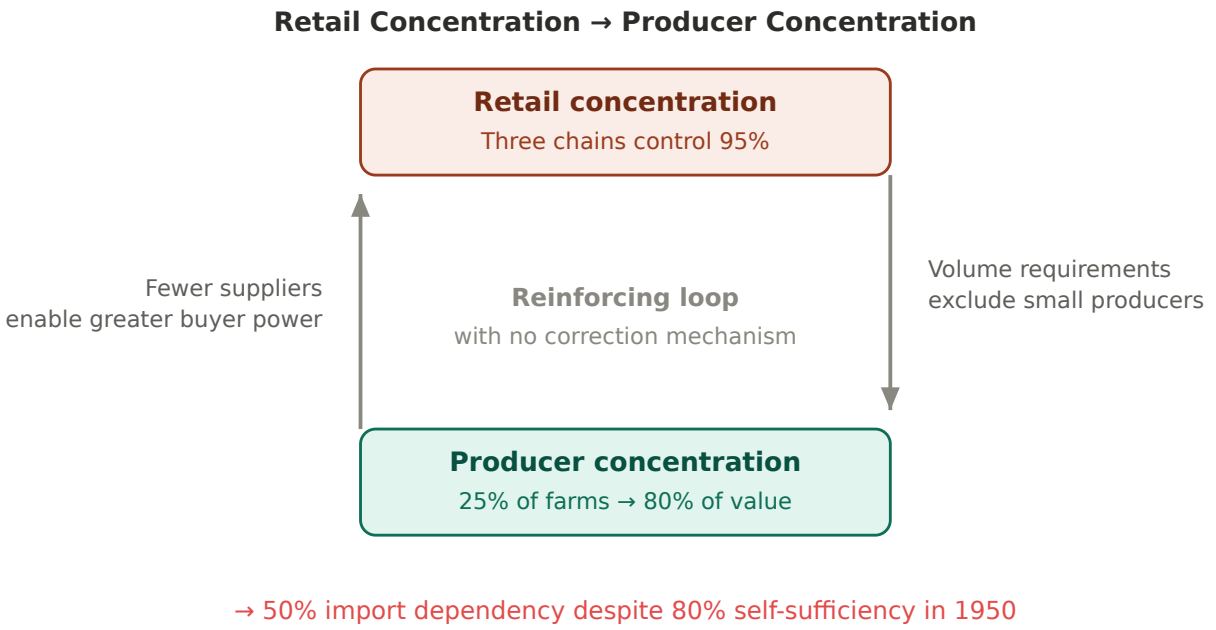
Evidence: Sweden's food self-sufficiency declined from 80% in 1950 to 50% today despite technological advancement. The system became more productive and less resilient simultaneously.

The Retail Oligopoly: Gatekeeping as Structural Vulnerability

The mechanism: Three chains (ICA, Coop, Axfood) control approximately 95% of grocery retail. This creates choke points where corporate procurement decisions determine what 10 million people can access, regardless of local production capacity or consumer preference.

The power law: Centralized distribution favors centralized production. Small-scale producers cannot meet volume requirements, quality standardization, or logistics coordination that retail oligopolies demand. The structure actively selects for concentration, making decentralization architecturally difficult rather than merely economically challenging.

The feedback loop: Retail concentration drives producer concentration (small farms cannot access markets) which drives further retail concentration (fewer suppliers enable greater buyer power). This is a reinforcing loop with no internal correction mechanism.



Just-in-Time Vulnerability: Optimization Against Resilience

The mechanism: Modern food logistics minimize inventory through just-in-time delivery, creating efficiency under normal conditions and catastrophic failure during disruption. Swedish cities maintain approximately 3 days of food inventory.

The trade-off: Just-in-time systems exchange resilience for efficiency. This optimization makes sense for predictable, stable systems. Food systems are neither—they depend on weather, geopolitics, energy prices, and biological processes. Optimizing for efficiency in inherently variable systems is architectural malpractice.

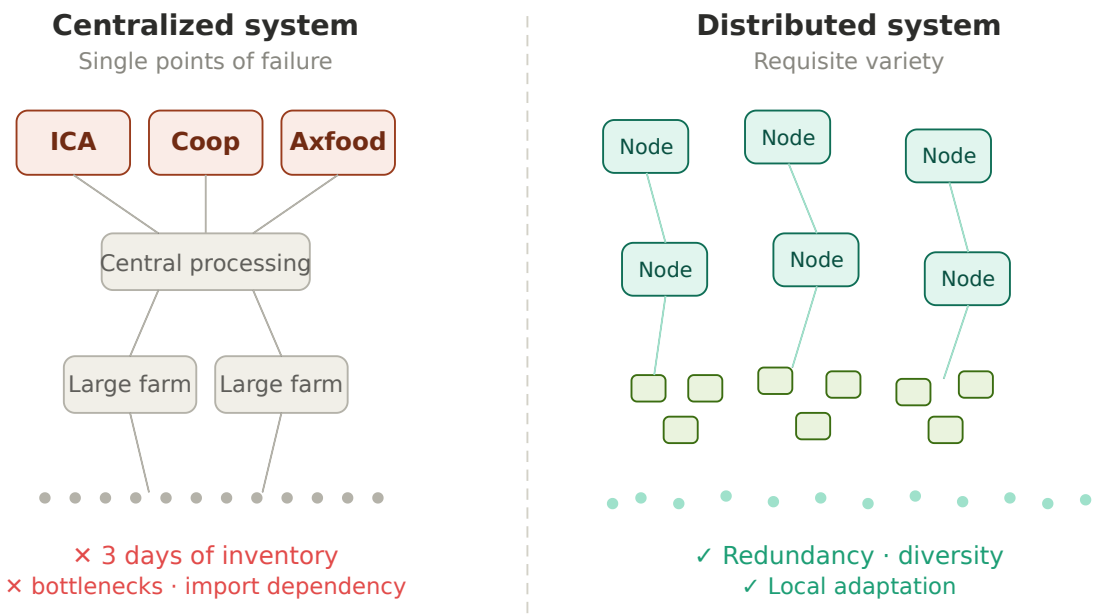
The evidence: COVID-19 demonstrated this fragility when supply chain disruptions created shortages despite adequate global production. The problem was not scarcity but distribution architecture.

Subsidy Misalignment: Funding the Wrong Controller

The mechanism: EU Common Agricultural Policy (CAP) funds (approximately 4.5 billion SEK annually to Sweden) flow predominantly toward large-scale operations, creating path dependencies that reinforce centralization. The subsidy structure rewards scale over diversity, volume over resilience.

The incentive problem: Farmers respond rationally to subsidy structures. When payments favor monoculture and scale, that is what emerges. The system is not broken—it is functioning exactly as designed. The design itself violates subsidiarity and resilience principles.

The alternative: Subsidy redirection toward bioregional diversity, small-scale operations, and regenerative practices would shift incentive structures, enabling rather than constraining distributed food systems.



Subsidiarity-Based Interventions

Municipal Procurement as Leverage Point

Swedish municipalities control massive food budgets for schools, hospitals, and elder care. Redirecting 25% toward local sourcing creates immediate market pull—without waiting for the national government. Ten pilot municipalities can start tomorrow.

Digital Infrastructure for Direct Contact

REKO rings show that Swedes want to buy directly from farmers—but Facebook groups don't scale. What if we build a federated, open platform that no chain can buy out? Where Swish and BankID form the backbone,

and every municipality can run its own node.

Read more: digital infrastructure and platforms →

Land Access and Workforce

The average age of Swedish farmers is 58. Only 8% are under 40. Meanwhile, new arrivals with agricultural experience are unemployed. How do we bring in a new generation? Through innovative leaseholds, apprenticeship systems with shared risk, and a national platform for seasonal work.

Read more: workforce, land, and skills →

Distributed Processing Infrastructure

Denmark has 30 mobile slaughterhouses. Sweden has zero—despite sharing EU regulations. Why? And what would it take to launch cooperative mills, fermentation hubs, and mobile units in Jämtland, Skåne, and Västra Götaland?

Read more: processing infrastructure →

The Role of Households: Demand Flexibility

During COVID, yeast and flour ran out—not because production failed, but because nobody could bake. A resilient system needs households that can cook from scratch, accept seasonal variation, and stock an emergency box with local produce.

Read more: households and food culture →

Implementation: From Theory to Practice

Phase 1: Municipal Autonomy Activation (1–3 years)

Procurement transformation: Ten municipalities establish 25% local sourcing targets for public institutions, creating immediate market pull that enables producer response.

Platform launch: Open-source infrastructure for direct sales, cooperative management, and supply chain coordination goes live in pilot regions, establishing proof of concept.

Processing infrastructure pilots: Mobile and cooperative processing units launch in three regions, demonstrating operational viability and establishing replicable models.

Phase 2: System Integration (3–7 years)

Subsidy redirection: Redirect 30% of CAP funds toward local food system development through mechanisms rewarding municipalities for increasing local production percentages.

Logistics coordination: Utilize existing rail and postal infrastructure for inter-regional food transport. Establish "food terminals" at medium-sized rail stations for last-mile distribution.

True cost accounting: Develop and implement comprehensive food system metrics including environmental and social costs.

Phase 3: Structural Transformation (7–15 years)

Educational integration: Incorporate food systems literacy and practical growing skills into curriculum at all levels.

Legal innovation: Create new legal entities optimized for food commons governance: cooperative-trust-foundation hybrids and bioregional food associations.

Resilience standards: Establish municipal food provisioning requirements in comprehensive planning.

The transition will not follow a straight line. Complex systems change through disruptions, experiments, and portfolio strategies—not through central planning.

Read more: transition dynamics and resistance →

The Swedish Advantage

Sweden possesses several characteristics that enable food system transformation:

Digital infrastructure: High connectivity and literacy enable platform-based coordination without centralized control.

Cooperative tradition: Folkrörelser provide cultural precedent and organizational experience for distributed coordination.

Municipal autonomy: Swedish constitutional arrangements already grant significant kommunal authority, enabling action without national permission.

Environmental consciousness: Cultural readiness for sustainability arguments creates political space for transformation.

Relative wealth: Resources exist for transition investment, reducing purely economic barriers.

Expected Resistance and Responses

"Lagom" as an Obstacle and Opportunity

The concern: The Swedish preference for balance and incremental change may slow down the transition.

The reframing: *Lagom*, correctly understood, is radical. The balanced, moderate system IS the regenerative, bioregional system. Industrial monoculture is the extremism that *lagom* should reject.

Arguments About Economic Efficiency

The concern: Small-scale farming appears uneconomical according to conventional metrics.

The response: Conventional metrics exclude externalities. Full cost accounting that includes environmental degradation, health impacts, and resilience value inverts the efficiency calculation.

"We Cannot Feed Ourselves" – Fatalism

The concern: The short growing season makes food sovereignty impossible.

The evidence: Sweden achieved 80% food self-sufficiency in 1950 with lower technology. The current dependence on imports is a result of political choices, not agricultural impossibility.

Institutional Path Dependencies

The concern: Trust in centralized authorities creates a wait-for-the-government attitude.

The activation: The Swedish tradition of civil society associations enables leadership from civil society where the government follows suit.

Co-optation – When the Establishment Absorbs the Change

The concern: ICA, Coop, and Axfood launch their own "local" labels, buy out alternative platforms, and lobby for watered-down definitions of locally produced. Then the incumbents win anyway.

The response: Cooperative ownership with limited dividend rights, federated architecture (no one can shut down the network), and a statutory definition of "local" in public procurement. Read more about co-optation resistance in the section on digital infrastructure.

Conclusion: Choose Between Apparent Efficiency and Real Resilience

Sweden's current food system is not efficient. It is **fragilely optimized**. It looks sophisticated—just-in-time logistics, centralized warehouses, three dominant chains—but it cannot withstand any major stress. An energy crisis, a war, a new pandemic, a crop failure in Central Europe: the system has no margins, no redundancy, no diversity.

The problem is not a lack of technology or bad farmers. The problem is an **architectural flaw**: we have built a system that optimizes for cost per unit under stable conditions—and ignores all other qualities (resilience, sovereignty, adaptability). It is like building a car without airbags to make it lighter and cheaper. It runs well—until it crashes.

The choice is simple, but not easy:

- **Continue as today**: More subsidies for large-scale monoculture, more centralized logistics, increased imports, continued depletion of the countryside and soils. Short-term cheap. Long-term suicide.
- **Start the transition now**: Municipal procurement as a driving force, open digital platforms, innovative leases, mobile slaughterhouses, training of new farmers, households that can cook from scratch. Short-term work. Long-term survival.

The Swedish head start is real: digital infrastructure, cooperative tradition, municipal self-government, environmental awareness, relative prosperity. We have all the prerequisites that other countries lack. The only thing missing is **the decision to begin**.

Three Things You Can Do Tomorrow

If you are a **municipal politician**: Propose a target of 25% local procurement by 2028. Use existing municipal autonomy—you do not need to ask the government for permission.

If you are a **farmer or horticulturist**: Team up with three neighbors and start a REKO group or a cooperative platform. Stop waiting for ICA to call.

If you are a **consumer**: Ask your school, your nursing home, your hospital where their food comes from. Buy directly from the farmer once a week. Teach a child to ferment cabbage or bake bread.

The Final Reason

A resilient food system is not a nostalgic dream of an agrarian society. It is a **survival strategy for the 2020s and beyond**. Climate change, geopolitical instability, energy insecurity—all three strike directly at the way we eat. A system that cannot withstand a disruption is not modern. It is outdated in a dangerous way.

Sweden has the capacity. We have the knowledge. We have the money (if we redistribute the subsidies). We have movements, associations, municipalities, farmers, and citizens who are ready.

What remains is to stop optimizing for vulnerability—and start building for resilience.

It is not a question of technology or economics. It is a question of stopping fooling ourselves that fragility is efficiency.

This analysis applies cybernetic principles and systems thinking to Swedish food systems. Data sources include Jordbruksverket, SCB, and international research on food system resilience and complexity.

Digital Infrastructure for Direct Contact

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Summary: REKO shows the demand exists—but Facebook groups don't scale. This section describes a federated, open platform built on Swish and BankID, with a cooperative ownership model that makes buyouts and co-optation structurally impossible.

The Measure

Open-source platforms that enable direct contact between producer and consumer, cooperative coordination, and transparent origin tracking. This breaks the retail oligopoly by creating alternative distribution channels—not by competing on price, but by offering relationships, traceability, and local adaptation.

Swedish Model – The REKO Ring

REKO (*Rättvis Konsumtion* / Fair Consumption) is a movement where consumers and producers meet via local Facebook groups. Orders are placed digitally, and delivery takes place at a shared time and location. REKO has already shown that Swedes want to buy directly—but the model is manual and scales poorly. Our measure builds on REKO's idea by providing **open and automated infrastructure** that handles payment, identity, logistics, and warehousing, without a platform owner being able to take control or change the rules.

Technical Architecture

Federated rather than centralized platforms. Each municipality or cooperative can run its own instance (like its own website), but all instances talk to each other via a shared, open protocol. This means that a farmer in Dalarna can sell to a consumer in Skåne without being connected to the same platform provider. Interoperability without central control—exactly how email works.

Onboarding Strategy – How Do We Get People to Use It?

1. Start with public procurement as the driving force: Municipalities that have already introduced 25% local procurement require their suppliers to use the platform for reporting and invoicing. This way, farmers aren't forced in—they are given a business reason.

2. Build on existing REKO groups: Offer group administrators a smoother transition from manual Facebook posts to automated orders. Many REKO producers are already technically curious—they will become early adopters.

3. Integrate with Swish and BankID: Payment takes place directly between the consumer and the producer (no platform holds the money). BankID guarantees that both parties are real people/companies, which builds security without an intermediary.

4. Municipal "food hubs" as physical nodes: In each pilot municipality, an existing location (a community center, library, or a farmer's yard) is designated where goods are collected and picked up. The platform shows the physical path of every product—from soil to hub.

Why Would Producers and Consumers Switch from ICA?

Not switch—complement. The platform does not claim to replace grocery retail for all goods. It creates a **parallel track** for local, seasonal, and traceable products that currently do not reach the market because they do not fit the volume and logistics requirements of retail. For the consumer: lower prices (no middleman), greater knowledge of origin, and a relationship with the farmer. For the producer: higher margins and immediate payment.

Risks and Responses to Criticism

- *"No one will bother with a new app."* – That is why we are building on Swish and BankID, which are already installed on every Swedish phone. The interface can be as simple as an SMS or a web form.
- *"ICA will copy the concept."* – Let them. A centralized platform from ICA can never offer the same transparency and local adaptation as a federated, open solution. A cooperative ownership model means that the incentives will not be to maximize extraction, but to minimize friction for the members.

Co-optation Resistance

ICA, Coop, and Axfood will not fight back openly. They will launch their own "local" labels, start platforms for direct sales, and use their market influence to neutralize the reform without changing the underlying structure.

The mechanism of co-optation:

1. **Symbolic adaptation** – The chains create their own "REKO-like" features in their apps, but dictate the terms (pricing, logistics, data extraction).
2. **Regulatory capture** – Industry organizations lobby to make the definition of "local food" so broad that even centrally produced goods qualify.

3. **Buyouts** – When an open platform begins to grow, the founders are offered a sum they cannot refuse.

How to build co-optation-resistant systems:

- **Ownership structure that locks in the mission**: Platforms and cooperatives should be organized as **economic associations with limited dividend rights** or as **foundations** where the bylaws require a two-thirds majority to change the purpose. This makes a buyout legally impossible or pointless.
- **Open standards, not open source alone**: Open source is not enough if the platform is run by a single company. The requirement is a **federated architecture** (like email) where no single actor can shut down or take over the network.
- **Statutory definition of "local"**: Preempt the industry's dilution of the concept by pushing through a legal definition (e.g., within 50 km or the same bioregion) in public procurement.
- **Member-owned, not user-owned**: Cooperative models with "one member – one vote" and the right of first refusal for members in the event of a sale make buyouts practically impossible.

Examples from Sweden: Lantmännen started as a cooperative but has over time become increasingly centralized and market-oriented. This shows that a cooperative form is not a guarantee—the bylaws must be specifically written to counteract co-optation. Lessons from the Swedish Tourist Association (STF) and the Cooperative Union (KF) during the 20th century show both potential and pitfalls.

Resistance will not primarily come as open conflict, but as an invitation to "collaboration" that hollows out the content of the reform. Therefore, every measure must specify not only *how* it is implemented, but also *what form of ownership and governance makes it co-optation-resistant*.

Integration With Other Measures

The platform is connected to mobile slaughterhouses, mills, and fermentation units (see processing infrastructure) so that small producers can coordinate logistics and capacity. It becomes the backbone of a distributed food system—not a new centralized marketplace.

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Workforce, Land, and Skills

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Summary: Only 1% of the workforce is in agriculture, but a distributed system requires a doubling. This section presents four concrete sources—new arrivals, lifestyle movers, automation as complement, and seasonal work—plus financing through redistribution of existing EU subsidies.

Land Access Through Leasehold Innovation

The intervention: Reform agricultural leasehold arrangements to enable new farmer entry. Potential mechanisms include carbon credit systems that incentivize landowners (Church of Sweden, Sveaskog, municipal land banks) to lease to regenerative practitioners.

The barrier removal: Current land prices create entry barriers that favor established operations. Leasehold arrangements reduce capital requirements while enabling experimentation and adaptation.

The succession solution: With average farmer age at 58 years and limited succession pathways, leasehold innovation addresses both entry barriers and demographic transition simultaneously.

Where Are the People?

The problem in numbers: The average age of Swedish farmers is 58. Only 8% are under 40. The share of the workforce employed in agriculture has decreased from over 20% in 1950 to just over 1% today. At the same time, a distributed, regenerative food system requires *more* human attention per hectare, not less. Monoculture with large machines gets by with few working hours per calorie. Diverse farming with animals, vegetables, cover crops, and direct sales requires knowledge, presence, and labor.

The paradox: The system has optimized away people—and then complains that there are no farmers left. The transition does not require everyone to become a farmer, but it does require a **doubling or tripling** of the workforce in primary production and local processing.

Four Realistic Sources

1. New Arrivals and Integration

For several decades, Sweden has received people with agricultural experience from countries where small-scale, labor-intensive farming is the norm. Today, unemployment among the foreign-born is high—at the same time as Swedish agriculture is crying out for labor. The connection is broken.

Measure: **Municipal "agriculture intro" programs** that combine SFI (Swedish for Immigrants) with internships on farms transitioning to regenerative methods. Model: Denmark's "landbrugsgrunduddannelse" (basic agricultural education) with a guaranteed internship. Financed through the Swedish Public Employment Service's establishment assignment and EU agricultural funds.

Example: Even today, there are examples like "Fair Work in Skåne" and "Grasp the Food" in Västra Götaland—but they are too small. Scale them up with municipal coordination.

2. Lifestyle Movers and "Returnees"

The pandemic showed that people can move from the city to the countryside if there are jobs, schools, and a sense of community. Many dream of working with food, but the threshold is high: land prices, capital requirements, solitary work, poor pay.

Measure: **Apprenticeship systems with shared risk** – Experienced regenerative farmers receive subsidies to take on apprentices. The apprentice receives a minimum wage the first year, then gradually more. The municipality provides supervisor training and venture capital (if the crop fails, the apprentice does not lose money).

Example: "Greenhorns" in the US, "Arbeitsgemeinschaft bäuerliche Landwirtschaft" in Germany. Sweden has no equivalent—a gap to fill.

3. Automation as a Complement, Not a Replacement

Many believe that robots will solve the labor shortage. This is partly true—for weeding, milking, and harvesting certain crops. But diverse systems (different crops, animals, processing methods) are difficult to automate. The point is to **use automation for the dull, heavy, repetitive tasks**—and free up humans for the qualified work: pruning, animal health, fermentation, and customer relations.

Measure: **Cooperative machinery pools** with light, smart implements (small electric tractors, row-weeding robots) shared by multiple farms. Financed through investment support for small-scale mechanization (today, such support goes almost exclusively to large machines).

4. Seasonal Work and Circular Labor

Swedish agriculture has always been seasonal. Today, seasonal workers come from Thailand, Ukraine, and other countries via complicated visa processes. At the same time, there are students, part-time retirees, and furloughed workers from other sectors who could work a few weeks a year—if there were an easy point of entry.

Measure: **A national platform for seasonal jobs** (built on the digital infrastructure) where farms post their needs ("3 weeks in August, berry picking") and workers book shifts with direct Swish payments. Simplified registration, no lengthy bureaucracy.

Skills – Not Just Hands

Labor without knowledge is worthless. A distributed system requires people to be able to read the landscape and soil, plan crop rotations and intercropping, ferment, preserve and dry, handle animals with low stress, and repair and build simple structures.

Measure: **High school programs in regenerative agriculture** (currently only a few exist, e.g., at Dalsland's Folk High School). Link this to the natural resource use program but with a focus on small-scale diversity, not large-scale monoculture. **Folk high school courses** for adults wanting to change careers—with guaranteed internships.

Where Is the Money?

Investments in labor and skills can be financed through the **redistribution of existing EU subsidies** (CAP), which today go toward area-based grants for large-scale farms. Every billion shifted from large-scale to small-scale creates more jobs per *krona*—because small-scale farming is more labor-intensive. This is not a cost, but a **redistribution of employment** from machines to people.

Objections and Responses

- *"No one wants to work in agriculture—it's poorly paid and grueling."* – That is why we must improve the conditions. Distributed systems can yield higher margins per hour (direct sales, processing) and a better working environment (less solitary work, shared machinery pools). This isn't the 1800s—it's the 2000s with technological support.
- *"Swedes are too expensive to hire in agriculture."* – Today, we import cheap labor from low-wage countries. This is unsustainable and exploitative. The transition means paying a fair price for food—which leaves room for Swedish wages.

- *"This is too slow—the crisis will hit before we have time to train people."* – That is why we combine the short term (seasonal platforms, apprenticeship systems) with the long term (high school programs). Preparedness is about parallel timescales.

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Distributed Processing Infrastructure

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Summary: Denmark has 30 mobile slaughterhouses—Sweden has zero, due to regulation rather than technology. This section presents three regional pilots (Jämtland, Skåne, Västra Götaland), the Danish model, concrete cost estimates, and the regulatory changes required.

The Measure

Mobile and cooperative processing units (slaughterhouses, mills, fermentation plants, dairies) that serve multiple small producers. This reduces the capital requirements that currently force consolidation and allows small-scale producers to process their raw materials locally instead of sending them to centralized facilities—or being forced to sell unprocessed goods at low prices.

Economies of scope: Instead of seeking large-scale economies of scale (which drive centralization), distributed processing achieves economies of scope—shared infrastructure serving varied production at an appropriate scale. A mobile slaughter unit can serve 20 farms within a 50-kilometer radius. A cooperative mill can grind grain from 30 growers. A shared fermentation plant can produce fermented vegetables, kombucha, or lactic acid products for several farms simultaneously.

Swedish Barriers – Why Doesn't It Already Exist?

Regulatory Barriers

- **Slaughterhouse regulations**: Swedish slaughterhouses must meet the same hygiene requirements regardless of scale. Mobile slaughterhouses are practically impossible because the rules require separate rooms for clean and unclean operations, which is difficult in a truck. Denmark has exemptions for small-scale mobile units—Sweden does not.
- **Dairy regulations**: Farm dairies are subject to the same control programs as Arla. The cost of certification and sampling makes small-scale cheese and yogurt production unprofitable.
- **Mills**: The EU hygiene regulation (EC 852/2004) is interpreted differently in different member states. In Sweden, a separate mill building with floor drains and stainless steel is often required—in Denmark, simpler solutions are permitted for direct sales.

Economic Barriers

- Investment subsidies from the EU go almost exclusively to large-scale equipment. A small mill for SEK 500,000 rarely receives support; a large mill for SEK 50 million does.
- Banks are unaccustomed to financing cooperative infrastructure. An individual farmer cannot get a loan for a mobile slaughter unit—but a cooperative with ten members has better prospects, although banks lack models for evaluating such projects.

Swedish Role Models and Exceptions

- **The Dalsland slaughterhouse** – a smaller, cooperative slaughterhouse that has survived through hard work, but struggles with regulations written for large-scale operations.
- **Klövermjölk farm dairy** (Skåne) – an exception that proves it is possible, but the founders testify to a three-year permit process and costs of over half a million SEK before the first liter was sold.
- **The REKO rings' own mill initiatives** – in several REKO groups, farmers have joined forces to buy a shared mill, but are forced to use it illegally or transport grain to a large mill in another region.

The Danish Model

Since 2015, Denmark has permitted mobile slaughterhouses for up to 500 animals per year per unit. The rules do not require separate rooms, just thorough cleaning between animals and documented hygiene routines. The result: over 30 mobile units in operation, serving thousands of small farms. Swedish adoption requires political will and regulatory changes—not technological development.

Proposed Pilot – Three Regions, Three Types

Region 1 (Jämtland) – Mobile Slaughter Unit

- **Technology**: Slaughter container with fold-down walls, cleaned between each animal. The same model as in Denmark.
- **Regulatory change**: The County Administrative Board grants a time-limited exemption. Evaluation after 2 years.
- **Organization**: Economic association with 15 farms sharing the investment (approx. SEK 1.5 million). State investment subsidy of 50%.

Region 2 (Skåne) – Cooperative Mill and Bakery

- **Technology**: Stone mill (SEK 300,000) and dough mixer (SEK 100,000) in a converted container. Rented out to growers and bakers.

- **Regulatory change**: The Swedish Food Agency issues a guiding decision that container solutions are approved for direct sales under 10 tons/year.
- **Organization**: Cooperative society where each member buys shares (SEK 5,000 each). Members get priority access to the mill and a discount.

Region 3 (Västra Götaland) – Fermentation Hub

- **Technology**: Fermentation vessels (100–500 liters), cold room, packaging line. Shared by 10 vegetable growers.
- **Regulatory change**: No new law is required, but the County Administrative Board recognizes the hub as a shared food premises (instead of each grower needing their own permit).
- **Organization**: Cooperative lease agreement—growers pay per hour used and per liter of fermented product.

Regulatory Changes Required

1. **Introduce a new category in the Food Act**: "Small-scale mobile unit" with simplified requirements (separate rooms are not required, just cleaning protocols).
2. **Change the EU subsidy criteria** so that 30% of investment support is earmarked for mobile and cooperative units under SEK 2 million.
3. **Create a national "green lane"** for permits—one administrator per county with specific expertise in small-scale, distributed processing, with a maximum processing time of 6 months.

Financing and Economics

- **Investment**: A mobile slaughter unit costs SEK 1.2–1.8 million. With a 50% investment subsidy, this becomes SEK 600,000–900,000 shared among 15 farms = SEK 40,000–60,000 per farm.
- **Operations**: A butcher at 50% employment (SEK 300,000/year) plus cleaning, administration, and transport. At 500 slaughters per year, the cost is about SEK 150 per slaughter + margin. This is competitive with large-scale slaughterhouses that charge SEK 100–200 for slaughter but require long transports.
- **Socio-economics**: Each mobile unit creates 1–2 local jobs, reduces animal transport (lower emissions, less stress), and enables small farms to survive. An evaluation of Danish mobile slaughterhouses showed a socio-economic return of 2.5 times the investment.

Objections and Answers

- *"This is too expensive—we cannot subsidize every little unit."* – The subsidy is a one-time investment, not ongoing. Danish experience shows that mobile units become self-sustaining after 2–3 years.

- *"Hygiene cannot be guaranteed in mobile units."* – The Danish model has worked for almost a decade with fewer remarks than stationary small slaughterhouses. The problem is Swedish regulatory conservatism, not technology.
- *"Why not just build more small stationary slaughterhouses?"* – Because mobility is crucial for serving sparsely populated areas. A stationary unit requires animals to be transported. The most expensive and stressful thing for the animals is precisely the transport.

Connection to Other Measures

- **Digital platforms** book time in mobile units, handle certificates, and process payments.
- **Workforce** – mobile units can be an entry point for young butchers and processing workers who otherwise would not get internships.
- **The role of households** – fermentation hubs can also offer courses in home canning and preserving.

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The Role of Households: Demand Flexibility and Food Culture

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Summary: Centralized systems create passive consumers. This section describes four measures for demand resilience—school kitchens as pedagogy, municipal seasonal calendars, preserving nodes via libraries, and emergency boxes with local produce.

The Problem No One Is Talking About

A resilient food system requires not just distributed production—it requires **demand flexibility**. Swedish households have grown accustomed to all vegetables being available year-round, meat coming in standardized plastic packaging, and cooking taking under 15 minutes. This convenience is a product of the centralized system—and an obstacle to transition. If a crisis hits and farmers are growing carrots and cabbage (which survive the Swedish winter) while consumers demand avocados and lettuce year-round, it won't matter how good the production is.

The mechanism: Centralized systems create passive consumers. When the system breaks down, there is no built-in ability to adapt demand. A distributed system requires active citizens who can cook from scratch, make use of seasonal harvests, and tolerate a fluctuating supply.

The proof: During COVID-19, we saw that food shortages occurred not because there was a lack of calories, but because consumers could not or would not adapt. Yeast and flour ran out—not because production failed, but because home baking became a trend the system wasn't built to handle.

Measures for Demand Resilience

1. School Kitchens as Pedagogical Tools

Sweden's schools serve millions of meals daily. Today, these kitchens are often locked into centralized contracts and pre-made food products. By **reintroducing basic cooking from raw ingredients** in school kitchens—and linking this to the Home and Consumer Studies curriculum—a generation can learn how to handle root vegetables, ferment vegetables, bake bread, and plan meals according to the season.

Example: Municipalities that have already introduced local procurement can require school kitchens to participate in educational initiatives. Cooking becomes not just catering, but pedagogy.

2. Seasonal Calendars and Public Kitchens as Role Models

Every municipality develops a **seasonal calendar** for public meals: in January, it's root vegetables and fermented cabbage; in July, it's berries and early vegetables. The same calendar is made available to households via apps and notice boards. When citizens see that the hospital, the school, and the nursing home are eating seasonally, it becomes normalized.

3. Support for Home Preserving and Processing

During the 20th century, there was systematic advice on home preserving through Rural Economy and Agricultural Societies and educational associations. Today, that knowledge is fragmented. Measure: **Municipal "preserving nodes"** where citizens can borrow equipment (steamers, drying cabinets, fermentation vessels) and receive guidance. Financed via educational associations or as part of library "workshop loans."

Example: Libraries in Umeå and Malmö already have "borrow a sewing machine" or "borrow a jam pot" programs. Scale this up and orient it toward food preparedness.

4. Home Emergency Preparedness – From Abstract to Concrete

The Swedish Civil Contingencies Agency (MSB) recommends that households have food for a week. But the recommendation is generic. Measure: **Link emergency preparedness to local production**. Each municipality develops an "emergency box" based on locally grown, dry, and fermented foods (oats, peas, root vegetables, fermented cabbage). Households are encouraged to build their own box with support from municipal information.

Why this isn't "nannying": It's not about moralizing over consumer choices. It's about building **systemic redundancy**—if the production chain is disrupted, there is a population capable of adapting. It is just as much infrastructure as a road or a bridge.

Integration With Other Measures

- **Digital platforms** can include seasonal calendars, recipes based on what is available locally, and booking systems for preserving nodes.
- **Distributed processing infrastructure** can offer communal preserving courses and access to equipment.

- Subsidy redirection (phase 2) can reward municipalities that integrate demand flexibility into their preparedness plans.

Objections and Responses

- *"Swedes will never accept eating seasonally."* – They accepted it until the 1970s. It is a matter of norms, not human nature. Norms can be changed through the example set by public institutions.
- *"This is small-scale and naive."* – On the contrary. A centralized system has **zero** built-in demand flexibility. Consciously designing for flexibility is a sophisticated system property, not naivety.
- *"People don't have time to cook from scratch."* – That is why we are not proposing that everyone become an artisan chef. We are proposing that the system should be capable of ensuring people **can** do it when needed. Just as most people have a fire extinguisher at home even though they don't use it every day.

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Transition Dynamics: How Systems Actually Change

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Summary: Complex systems don't change linearly. This section describes five strategies for the transition—safe-to-fail experiments, disruptions as accelerators, portfolio diversification, scale-crossing between levels, and the art of not acting too early.

The three phases in the implementation plan are a logical sequence, but the transition will not follow a straight line. Complex adaptive systems (like food systems) change through **disruptions, experiments, and portfolio strategies**—not through central planning. Therefore, we need a conscious design for the dynamics of the transition.

1. Safe-to-Fail Experiments – Not Fail-Safe

The principle: In a complex system, you cannot predict what will work. Therefore, you should conduct many small experiments that **can fail without the system collapsing**. This is the opposite of large-scale pilots that are "too big to fail."

Swedish application:

- **Ten pilot municipalities** in phase 1 are already safe-to-fail—if three fail, the other seven continue. It is important to evaluate openly and share failures, not just successes.
- **Mobile slaughterhouses** in three regions—different technical solutions (slaughter truck, container solution, cooperative facility). Let the regions choose differently, evaluate after two years, and scale up what works.
- **Digital platforms**—allow multiple parallel codebases (different open-source projects) that compete and learn from each other. No "national platform" that locks everyone in.

What it requires: Political and administrative tolerance for failure. Municipalities must dare to report "this didn't work" without being punished. A possible model: **trial operations under Section 24 of the Local Government Act**, which already allows deviations from laws in limited projects.

2. Disruptions as Accelerators

The principle: Crises (pandemics, war, energy price shocks, extreme weather) are not just threats—they are **windows for system change**. A resilient system utilizes disruptions to accelerate the transition, rather than just restoring the old.

Swedish application:

- **Link emergency preparedness to the transition**. When the next crisis hits (presumably energy prices or crop failures in Europe), use the political momentum to temporarily allow exemptions from rules that hinder distribution (e.g., slaughter rules for mobile units, direct sales without certification requirements in an emergency).
- **Preventative scenario training** – County Administrative Boards and the Swedish Civil Contingencies Agency (MSB) invite municipalities, farmers, and cooperatives to "stress tests": "What happens if Ukraine shuts down its grain exports? What happens if the price of diesel quadruples?". Based on the answers, the most vulnerable rules are identified—and then abolished in advance, not in a panic.
- **"Emergency boxes"** – use an actual disruption to test them. If a storm stops deliveries for a week, activate the emergency boxes in two municipalities and evaluate.

Example from Sweden: During COVID-19, the government temporarily allowed farm-level slaughter and direct sales of meat. It worked. The rules were reintroduced after the crisis. Next time, they should be made permanent.

3. Portfolio Strategy – Not "One Size Fits All"

The principle: Uncertainty requires diversification not only in production but also in strategy. A portfolio of different initiatives (technology, organization, policy) survives better than a single effort.

Swedish application:

- **Three parallel tracks**:
  - *Municipal track*: Procurement, land, planning.
  - *Cooperative track*: Farmers joining together in new forms (economic associations, co-owned infrastructure).
  - *Civil society track*: REKO groups, farming networks, consumer cooperatives.
- These tracks do not need to be synchronized. They can develop at different paces and cross-pollinate each other. If one track fails (e.g., municipalities backtrack after an election), the others can continue.

Metrics for the portfolio: Not just "share of local food," but also the **number of active initiatives, number of municipalities with preparedness plans, number of new farmers under 40, number of cooperative units**. A portfolio is measured on diversity, not on the maximization of a single variable.

4. Panarchy and Scale-Crossing

The principle: In complex systems, change occurs when an innovation "jumps" from one scale level to another—from a farm to a municipality, from a municipality to a region, from a region to the nation.

Swedish application:

- **From farm to village**: A farmer succeeds with direct sales. Neighbors see it and join in. Soon the village has its own food node.
- **From village to municipality**: The municipality sees the success and adapts its procurement. Now more farmers get access to public meals.
- **From municipality to region**: Several municipalities in the same region coordinate their procurement requirements, creating scale without centralization.
- **From region to nation**: The government sees that the regions have reduced their dependence on imports and changes subsidy rules after the fact (not before).

Practical measure: **An annual "scale-crossing workshop"** where actors from different levels (farmer, municipal director, county governor, agency) meet and identify bottlenecks for scale-crossing. The workshop produces a concrete list of rules that should be changed—and sends it to the responsible ministry.

5. Negative Capacities: Not Acting Too Early

The principle: In complex systems, it is sometimes best to **wait and observe** instead of acting. Introducing a national solution too early can lock in a subpar design.

Swedish application:

- **Delay national regulations** of platforms, direct sales, and slaughterhouses—let the phase 1 experiments run for three years before drawing conclusions.
- **Avoid "one size fits all" standards** for organic certification, traceability, and the working environment. Allow regional and cooperative variants.
- **Use sunset clauses**—all temporary exemptions are written with an automatic expiration date, but can be extended if they work.

Integration With the Phases

- **Phase 1** (1–3 years): Focus on safe-to-fail experiments and building a portfolio. Allow failures. Gather data.
- **Between phase 1 and phase 2** (years 2–4): Evaluate the portfolio. Identify which experiments succeeded. Use scale-crossing workshops to move successes upwards.

- **Phase 2** (3–7 years): Scale up what has worked, but retain diversity. Introduce permanent rule changes based on evidence from phase 1.
- **Phase 3** (7–15 years): The system now has multiple parallel structures (central, regional, local). The transition dynamics continue—new disruptions will come, new experiments will be needed.

Objections and Responses

- *"This sounds like unplanned experimentalism—we need decisive action, not academic fluff."* – On the contrary. Planned experimentalism is the only strategy that works in complex systems. Pretending that we can predict everything is naive and leads to an overreliance on central planning.
- *"Who coordinates all these experiments?"* – No one. Coordination arises through shared infrastructure (digital platforms, shared evaluation criteria), not through a chain of command. That is the point.
- *"How do we avoid failures damaging trust in the entire transition?"* – By **communicating failures as learning**. Every pilot municipality commits to publicly reporting both successes and failures in a standardized template. The role of politics is to protect the experiments from being prematurely condemned.

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This analysis is part of Swedish Subsidiarity's work toward a more distributed and resilient Sweden.
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